

Statistical and Computational Models

Lecture 4: Assessing models

Daniel de Kadt

Department of Government, Cornell University · Fall 2026

A quick recap

1. **Loss:** $L(y, \hat{y})$ prices prediction errors. We've been working with squared loss; its average over the DGP is the **EPE** – the number we want to minimize.
2. Two candidate models:
 - OLS regression (lecture 2) – assume a form, estimate β .
 - k -NN regression (lecture 3) – average the k nearest observations.

OLS prefers stability (low variance) at the cost of bias, while k -NN may track the true f more closely, at the cost of variance.

How do we characterise that trade off?

Roadmap

The one sample problem

The validation split

Cross-validation

The test set

The missing test curve

Previously we were able to 'see' a test curve because we were simulating data.

In reality we typically only have one draw of our data. Call this $\mathcal{D}$:

$$\mathcal{D} = \left\{ (\mathbf{x}_i, y_i) \right\}_{i=1}^n$$

So which observations can score the fit? None of them! Every observation we have is a training observation.

Test MSE needs observations on which the model was not fit.

The way out is one we already know: **hold out**. Fit on some rows, score on the others.

Refining the training MSE

Better approaches to estimating the EPE, using $\mathcal{D}$ alone:

1. Validation set approach.
2. Cross-validation.
3. Test set approach (ideal).
4. 'In-sample' adjustment – next lecture.

Roadmap

The one sample problem

The validation split

Cross-validation

The test set

Method 1: Validation set

Method 1: use a validation set

1. Randomly divide $\mathcal{D}$ into two sets, $\mathcal{I}_{\text{tr}}$ and $\mathcal{I}_{\text{va}}$ – typically half each.
2. Fit on the training set; estimate the error on the validation set.

Calculate the loss on the **validation set**:

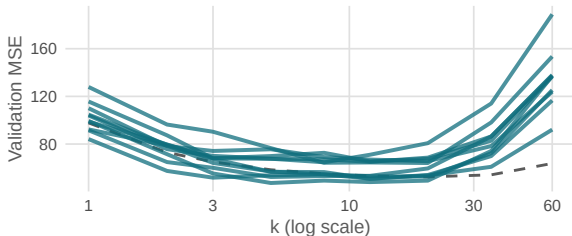
$$\text{MSE}_{\text{val}} = \frac{1}{|\mathcal{I}_{\text{va}}|} \sum_{i \in \mathcal{I}_{\text{va}}} (y_i - \hat{f}_{\text{tr}}(\mathbf{x}_i))^2$$



But this approach is very unstable. **Why?**

It can also run conservative, overestimating the error. **Why?**

The validation split in practice



One split of the 200 observations – dashed is the $\mathbb{E}[\text{EPE}]$ from the DGP. Let's try 9 more splits of the same observations.

At $k = 20$ the MSE runs from 49.3 to 80.7 – instability. They are also systematically high compared to the 'truth', overestimating the error.

They also pick poorly: 7 of 10 put their minimum below $k = 20$, which we know to be the optimal k .

Roadmap

The one sample problem

The validation split

Cross-validation

The test set

Method 2: Leave-one-out cross-validation

Method 2a: leave-one-out cross-validation

For each $i = 1, \dots, n$:

1. Move observation i to the validation set.
2. Fit on the remaining $n - 1$ observations.
3. Score observation i .

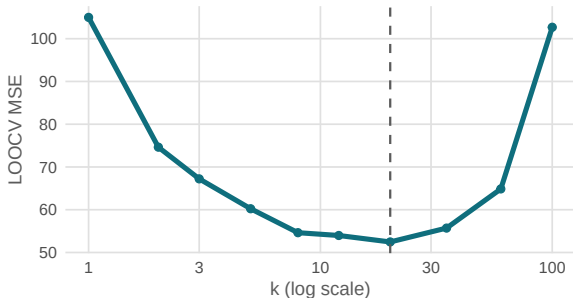
Average the n losses:

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}_{-i}(\mathbf{x}_i))^2$$



Each fit now sees $n - 1$ observations, and the power of chance is dramatically reduced. This solves both of our previous problems. **Why?**

LOOCV in practice



Hold out the orange observation; fit on the other 199. Another observation takes its turn. And another – all 200 get scored, at every k .

Average the 200 errors at each k to generate the LOOCV curve. Its minimum sits at $k = 20$ (nice).

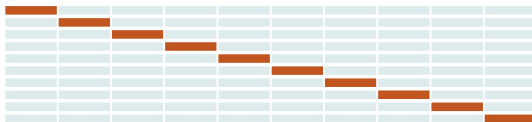
Method 2: K -fold cross-validation

Method 2b: K -fold cross-validation

1. Divide $\mathcal{D}$ into K **olds**: $\mathcal{I}_1, \dots, \mathcal{I}_K$.
2. For each fold j : fit on the other $K - 1$ folds, score fold j , call its loss MSE_j .

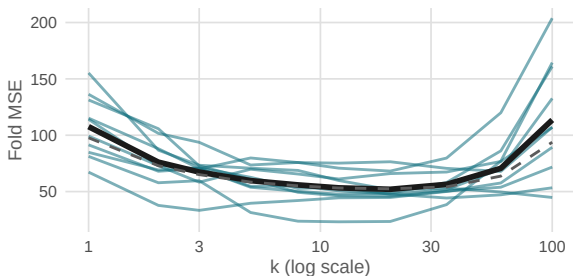
Average the K fold errors:

$$\text{CV}_{(K)} = \frac{1}{K} \sum_{j=1}^K \text{MSE}_j$$



Note: capital K counts folds; lower-case k still belongs to the neighbours. LOOCV is $K = n$; the validation split was $K = 2$ with no rotation. So what should K be?

K-fold CV in practice



With 200 observations and K folds, each fold has 20 observations, scored by a fit on the other 180. Consider one fold across values of k . We can then do this for all 10 folds. Averaging the folds at each k gives the black curve, $CV_{(K=10)}$, with the minimum of $k = 20$ (nice, again).

Dashed is the DGP $\mathbb{E}[EPE]$ – the CV still runs a little high due to the sample size sacrifice.

K-fold CV machinery

```
set.seed(6039)
K      <- 10
fold <- sample(rep(1:K, length.out = n))
mse  <- matrix(0, K, length(ks4))
for (j in 1:K) {
  tr <- train[fold != j, ]           # nine folds
  va <- train[fold == j, ]           # the tenth
  mse[j, ] <- colMeans((va$y - knn_all(tr, va$x)[, ks4])^2)
}
cv <- colMeans(mse)
ks4[which.min(cv)]
```

Note that the fold draw is part of the analysis – it happens once at the top.

Whatever our chosen loss is what we calculate across units. If the loss is wrong for the problem, CV optimises the wrong thing.

Choosing K

Bias:

- With few folds each fit sees far fewer than n observations, so $CV_{(K)}$ runs high.
- How much higher depends on whether the estimator is still improving between $n(K - 1)/K$ observations and n . Here, with 10 folds, hardly at all.

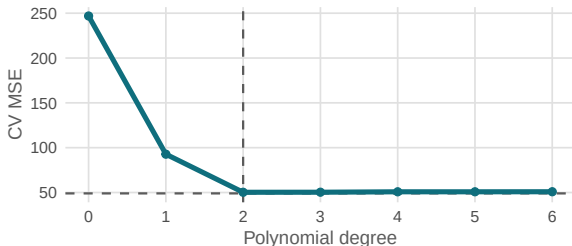
Variance:

- At $K = n$ the fits share almost all their training observations, so their errors rise and fall together. Averaging **correlated** errors removes far less noise than averaging independent ones.

$CV_{(n)}$ – LOOCV – carries the least bias and the most variance; $CV_{(2)}$ – the validation split, rotated once – does the opposite.

Cross-validation more broadly

We can use cross-validation to evaluate many things, it does not need to be k -NN. For example, we can use it to choose the polynomial degree for OLS.



CV picks $d = 2$; anything more flexible than $d = 2$ sits at the irreducible error floor, $\sigma^2 = 49$.

The CV standard error

$CV_{(K)}$ averages K fold errors, and thus carries a **standard error**:

$$\widehat{se}(CV_{(K)}) = \frac{sd(MSE_1, \dots, MSE_K)}{\sqrt{K}}$$

Read: how far we expect $CV_{(K)}$ to move if the folds were drawn repeatedly.

In our case, if we redraw the folds 50 times, $k = 20$ wins 49 times. The minimum stays between 51.4 and 55.9.

Cross-validation done wrong

Suppose our data carries 500 extra columns – none related to turnout, though we do not know that.

Temptation:

1. **Screen**: keep the 20 columns most correlated with y , using all 200 observations.
2. Fit OLS on **age** plus those 20 columns.
3. Cross-validate to estimate the EPE.

Is this a correct application of cross-validation?

Cross-validation done wrong

Design	CV MSE
Age only	92.7
The workflow: screen once, then CV	89.3
Screen inside each fold instead	130.9

Age only cross-validates at 92.7.

The proposed workflow lands at 89.3, which **looks like an improvement**.

Move the screening **inside** each fold and instead we get 130.9. The additional columns only hurt, because they are noise.

Question: why do the last two disagree? The fold must redo **every** data-dependent step: screening, standardising, imputing, choosing k . Failing to do this caused “leakage”.

Roadmap

The one sample problem

The validation split

Cross-validation

The test set

Method 3: Test set

Method 3 (a common refinement): use a test set

1. Randomly divide $\mathcal{D}$ into three sets: training, validation, and test.
2. Fit on the training set, choosing the model with CV or the validation set.
3. Once the model is chosen, score it on the test set, just **once**.



This estimates the EPE well if...

- the full sample is large, and randomly drawn
- the test set is itself large
- the test set was never touched before the test

This requires a lot of discipline. You need lots of data, you sacrifice data, and you lose researcher degrees of freedom.

Two tasks

Model selection: build the best possible model – choosing its shape or some tuning parameters.

Model assessment: how good is our chosen model?

References

- James, G., Witten, D., Hastie, T. & Tibshirani, R. (2021) *An Introduction to Statistical Learning*, 2nd ed. – **5.1.** statlearning.com.
- Lundberg, I., Johnson, R. & Stewart, B. M. (2021) "What Is Your Estimand? Defining the Target Quantity Connects Statistical Evidence to Theory." *American Sociological Review* 86(3): 532–565.
- Hastie, T., Tibshirani, R. & Friedman, J. (2009) *The Elements of Statistical Learning*, 2nd ed. – 7.10, 7.12.